

B.Tech. DEGREE MODEL EXAMINATION, FEB 2023.

First Semester

MATHEMATICS – I

(Common to All branches)

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

Section:

Date:

PART A – (10 × 2 = 20 marks)

Answer ALL Questions

All questions carry equal mark

1. Find the radius of curvature of the curve $y = e^x$ at (0, 1).
2. Define Evolute and Involute.
3. $x = r \cos \theta$; $y = r \sin \theta$, evaluate $\frac{\partial(x,y)}{\partial(r,\theta)}$
4. State chain rule for total derivative of $u(x, y)$ where $x = f(t)$ and $y = g(t)$.
5. Evaluate $\int_0^1 \int_2^5 xy \, dx \, dy$
6. Define area in polar co-ordinates
7. Write the general solution of $p^2 - 7p + 12 = 0$.
8. Solve: $(y - px)(p - 1) = p$.
9. Find the P.I of $\frac{d^2y}{dx^2} + y = x$.
10. Solve: $(D^3 - 12D + 16)y = 0$.

PART B – (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full question from each unit

UNIT - I

- 11.a) Find the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

11. b) Prove that the radius of curvature for the curve $x = a(\theta + \sin\theta)$; $y = a(1 - \cos\theta)$ is $4a \cos \frac{\theta}{2}$.

(OR)

- 12.a) Prove that $\beta(m, n) = \frac{\sqrt{m}\sqrt{n}}{\sqrt{m+n}}$, $m \& n > 0$

12. b) Find the circle of Curvature of the curve, $\sqrt{x} + \sqrt{y} = \sqrt{1}$ at $(\frac{1}{4}, \frac{1}{4})$

UNIT - II

13. Find the volume of the largest rectangular solid which can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

(OR)

- 14.a) Expand $xy^2 + 2x - 3y$ in powers of $(x + 2)$ and $(y - 1)$ upto the third degree terms.

- 14.b) If $u = \sin^{-1} \left[\frac{x+y}{\sqrt{x}+\sqrt{y}} \right]$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$.

15. Find the volume portion of the Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, which lies in first octant.

(OR)

16. Evaluate $\int_0^3 \int_1^{\sqrt{4-y}} dx \, dy$ by changing the order of integration.

UNIT - IV

- 17.a) Solve $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$

- 17.b) Find the orthogonal trajectories of the family of the parabolas $y^2 = 4ax$.

(OR)

- 18.a) Solve $p^2 + 2py \cot x = y^2$

18.b) Solve $y = (x - a)p - p^2$

UNIT - V

19.a) Solve using method of variation of parameter

$$\frac{d^2y}{dx^2} + 4x \frac{dy}{dx} = 4 \tan 2x$$

19.b) Solve $x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + 2y = \sin(\log x)$

(OR)

20.a) Solve $\frac{dx}{dt} + 2x - 3y = t$; $\frac{dy}{dt} - 3x + 2y = e^{2t}$.

20.b) Solve $(D^2 - 4D + 4)y = \cos h 2x + 3$.